The output effect of price discrimination revisited

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Introduction

- Classic problem of the output and welfare effects of third-degree price discrimination by monopolist
- Robinson (1933), Varian (1985), ACV (2010)

- Related literatures on second-degree price discrimination
- ... and on price discrimination in competitive settings

- Most of the literature has independent demands, and just two products
- Can more be said about the classic problem, especially when *demands are interdependent*?
- Analysis will focus on the total demand function $X(p)$ rather than individual product demands

Some results from ACV (2010)

- Focus on relative curvatures of demand

PROPOSITION 4: (i) *If demand is concave in the strong market and less concave, or convex, in the weak market then output increases with discrimination.* (ii) *If demand is convex in the strong market and concave, or less convex, in the weak market then output decreases with discrimination.* (iii) *If inverse demands are convex, and more so in the weak market, then output increases with discrimination, while* (iv) *if inverse demands are concave, and more so in the weak market, output falls with discrimination.*

- Constant elasticity result

PROPOSITION 6. *When demand functions have constant elasticities:* (i) *total output is higher with discrimination, and* (ii) *social welfare is lower with discrimination if the difference between the elasticities is at most* 1.

A motivating example

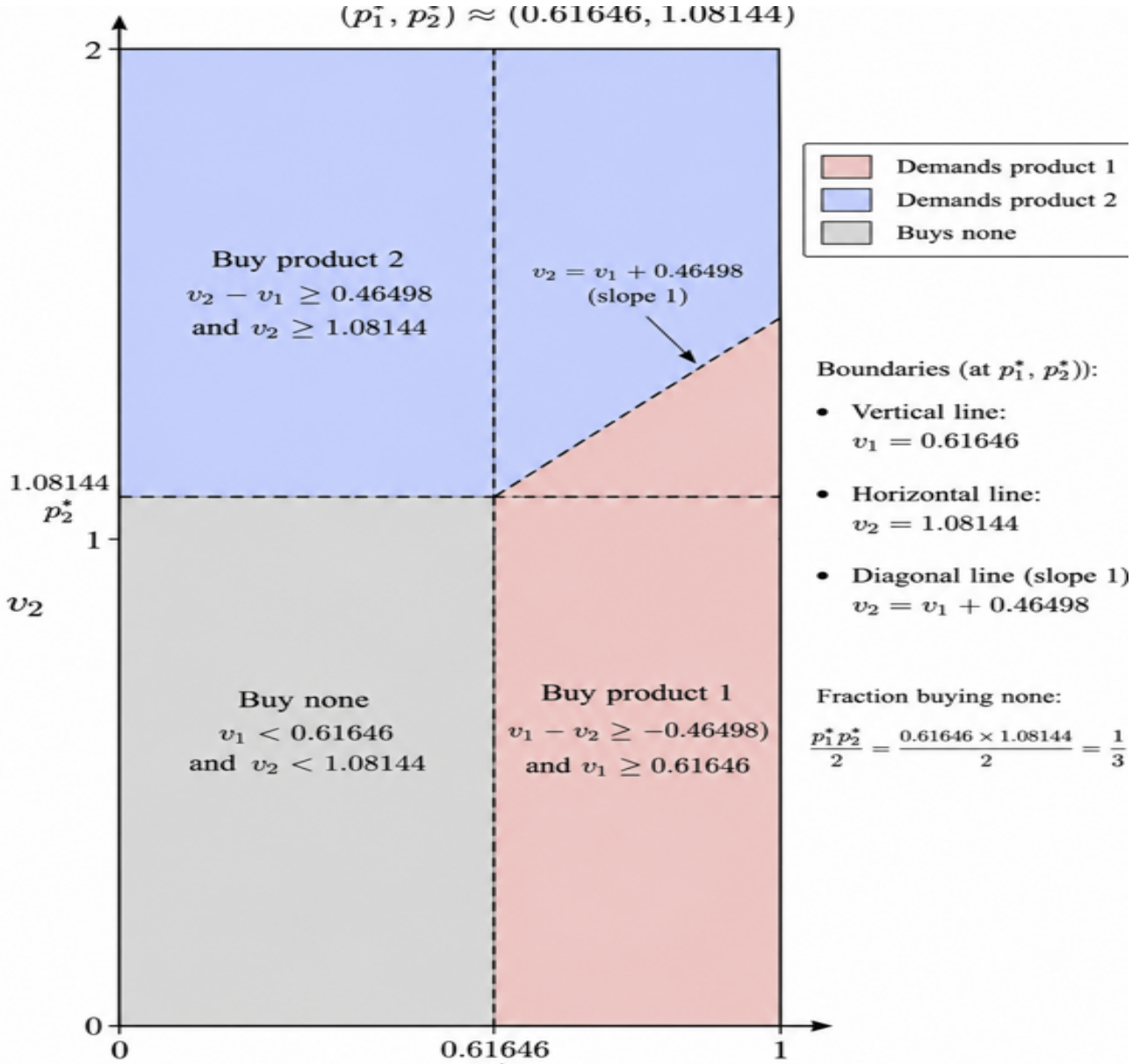
- Discrete choice with two products
- Unit cost $c \in [0, 1]$
- Values distributed uniformly and independently: v_1 on $[0, 1]$ and v_2 on $[0, 2]$

- Total demand

$$X(p_1, p_2) = 1 - \frac{1}{2} p_1 p_2$$

  - is homothetic in prices
  - and quasiconvex

Graphical illustration of product demands



Solution of example

- Let X^D and X^U denote total output with Discrimination and Uniform pricing respectively

- X^U drops sharply as c rises through 0.276, as the firm stops supplying a product.

- $X^U = X^D$ at $c = 0$ and $c = 1$
- $X^U < X^D$ for $0.276 < c < 1$, when just one product is supplied with uniform pricing
- $X^U > X^D$ for $0 < c < 0.276$, when both products are supplied with uniform pricing

- This last inequality can be related to $X(p)$ being **homothetic** and **quasiconvex**

A general "average output" lemma

- One way to compare "average outputs" is by allowing unit cost to vary
- Suppose that c is uniformly distributed on $[a, b]$ with $a < b$
- Profit $\pi(c)$ as a function of cost satisfies $\pi'(c) = -X(c)$ by the envelope theorem
- So expected output

$$\begin{aligned} \mathbb{E}[X] &= \frac{1}{b-a}\int_a^b X(c)dc = \frac{-1}{b-a}\int_a^b \pi'(c)dc \\ &= \frac{\pi(a)-\pi(b)}{b-a}. \end{aligned}$$

Corollary of lemma

- If $\pi^D(b) = \pi^U(b)$, so that profit is the same in the two regimes at cost b, but $\pi^D(a) > \pi^U(a)$, then

$$\mathbb{E}[X^D] > \mathbb{E}[X^U]$$

  and average output is greater with discrimination
- Thus, in the example, with c uniform on $[0, 1]$

$$\mathbb{E}[X^D] - \mathbb{E}[X^U] = \pi^D(0) - \pi^U(0) = 0.063 \ .$$

- But what can we say if cost is such that all products are in positive supply in both pricing regimes?

The total demand function tells us about individual demands

- The following analysis mainly uses $X(p)$
- Individual product demands $x_k(p)$ can be derived from $X(p)$ by

$$x_k(p) = -\int_0^\infty \frac{\partial}{\partial p_k} X(p_1 + t, ..., p_n + t)dt$$

  using Roy's identity and Slutsky symmetry

- See Armstrong & Vickers *JET* (2015) Lemma 1

The model

- A profit-maximizing monopolist supplies n products
- Price vector is $p = (p_1, ..., p_n)$
- Quantity vector is $x = (x_1, ..., x_n)$
- Total demand function is $X(p) = \sum_k x_k(p)$
- Cost $C(X)$ depends on aggregate output, e.g. $C(X) = cX$

Marginal revenue comparison

- Let $R^D(X)$ be maximum revenue with output X when discrimination is allowed, and let $R^U(X)$ be maximum revenue with X when uniform pricing is required
- $R^D(X) \geq R^U(X)$ with strict inequality except when uniform pricing is optimal
- Let $MR^D(X)$ and $MR^U(X)$ be the associated marginal revenues: $MR(X) = R'(X)$, assumed decreasing

- Key issue is the sign of $MR^D(X) - MR^U(X)$
- If $MR^D(X) > MR^U(X)$ the monopolist will supply more if it can discriminate
- If $MR^D(X) < MR^U(X)$ it will supply less if it can discriminate, and welfare will be lower

- Aim: express $MR(X)$ in terms of two price-sensitivity terms that enable comparisons

The elasticity of total demand

- Define
$$\eta(p) = -\frac{p \cdot \nabla X(p)}{X(p)}$$
as the *elasticity of total demand with respect to prices*

- In the special case of homothetic $X(p) = Y(h(p))$ with $h(p)$ HD1, we have that
$$\eta(p) = [p \cdot \nabla h(p)] \times \left(\frac{-Y'(h)}{Y(h)} \right) = -\frac{hY'(h)}{Y(h)}$$
is a function of h and hence of X.

The quasiconcavity/convexity condition

- Given p, let $p^0(p) = (p_0, ..., p_0)$ be the uniform price vector such that $X(p^0) = X(p)$
- Define the price-sensitivity term

$$\beta(p) = -\frac{p^0 \cdot \nabla X(p)}{X(p)} \ .$$

- Note that $\beta(p^0) = \eta(p^0)$

- Observe that

$$\beta(p) - \eta(p) \ \text{ has the sign of } \ (p - p^0) \cdot \nabla X(p) \ .$$

and therefore, with $X(p^0) = X(p)$:

- $\beta(p) \geq \eta(p)$ if $X(p)$ is quasiconvex, and
- $\beta(p) \leq \eta(p)$ if $X(p)$ is quasiconcave.

Expression for marginal revenue

- Define
$$\mu(p) \equiv \frac{\eta(p) - 1}{\beta(p)} \ .$$
- Let p^* be revenue-maximizing prices given some X
- Then
$$\begin{aligned} MR^D(X) &= p_0 \mu(p^*) \\ MR^U(X) &= p_0 \mu(p^0) \end{aligned}$$
- Thus $MR^D(X) - MR^U(X)$ has the sign of
$$\mu(p^*) - \mu(p^0) \equiv \left(\frac{\eta(p^*) - 1}{\beta(p^*)} \right) - \left(\frac{\eta(p^0) - 1}{\eta(p^0)} \right) \ .$$

Results with homothetic demands

- With homotheticity of $X(p)$, what matters is quasiconvexity/quasiconcavity
- With linear (including interdependent) demands, and all products supplied, we have homotheticity + quasiconvexity + quasiconcavity, so for given X

$$\begin{aligned} \eta(p^*) \quad &= \quad \eta(p^0) = \beta(p^0) = \beta(p^*) \\ &\Longrightarrow \quad \mu(p^*) = \mu(p^0) \end{aligned}$$

  and $MR^D(X) = MR^U(X)$ as expected
- In our example, which can easily be extended to n products,

$$\begin{aligned} \eta(p^*) \quad &= \quad \eta(p^0) = \beta(p^0) < \beta(p^*) \\ &\Longrightarrow \quad \mu(p^*) < \mu(p^0) \end{aligned}$$

  so $MR^D(X) < MR^U(X)$ for positive marginal revenue
- (In the example with $c = 0$ we have $\eta(p^*) = \eta(p^0) = 1$ with profit-maximizing pricing, so $\beta(p^*)$ does not feature and $MR^D = MR^U$.)

General comparative result

- If $\eta(p^*) \geq \max\{\eta(p^0), \beta(p^*)\}$ then $MR^D(X) \geq MR^U(X)$ and total output is greater with discrimination
- If $\eta(p^*) \leq \min\{\eta(p^0), \beta(p^*)\}$ then $MR^D(X) \leq MR^U(X)$ and total output is greater with uniform pricing. So is welfare.

The case of constant beta

- Consider the example with individual demands $x_k(p_k) = a_k - b_k e^{p_k}$, where

$$X(p) = a - \sum_k b_k e^{p_k}$$

  has $X(p)$ concave, so quasiconcave
- Then

$$\beta(p) = p_0 \left(\frac{\sum_k b_k e^{p_k}}{X} \right) = p_0 \left(\frac{a - X}{X} \right) .$$

- So $\eta(p^*) \geq \beta(p^*) = \beta(p^0) = \eta(p^0)$ and therefore $MR^D \geq MR^U$, so output is greater with discrimination

- Footnote: $\beta(p)$ is constant on level sets of X if $X(p) = Z(s)$, where $s \equiv \bar{p} + \phi(d)$ with $\bar{p} = \frac{1}{n} \sum p_k$ and d is a vector of price differences, so that $\beta(p) = -p_0 Z'(s)/Z(s)$.

Settings with underlying symmetry

- Previous examples had $\eta(p)$ or $\beta(p)$ constant on level sets of X
- More generally, when does uniform price vector p^0 maximize or minimize $\mu(p)$ subject to $X(p) = X(p^0)$?
- Generalizing the initial example, total demand

$$X(p) = 1 - \prod \gamma_k [F(p_k)]^{\alpha_k}$$

  yields results using weighted Jensen inequality arguments
- Thus with $y(P) \equiv \log F(P)$, uniform pricing maximizes MR given X if

$$(P - p_0\mu)\frac{y''(P)}{y'(P)} \quad \text{is decreasing in } P$$

  in the relevant range of μ.

A complementary argument for constant elasticities

- ACV (2010) showed that output is greater with discrimination with constant elasticity demands
- This has $\beta(p^*) > \eta(p^*) > \eta(p^0)$ so the general comparative result does not apply
- But we can use another way of expressing MR. Define

$$\lambda(p) \equiv \frac{\sum_k (\eta_k - 1) p_k x_k}{p_0 X \eta(p)}$$

- Then $\mu(p) = \lambda(p)$ at both $p = p^0$ and $p = p^*$
- With independent demands (and under other conditions more generally)

$$\lambda(p^*) \geq \frac{\tilde{\eta} - 1}{\tilde{\eta}} \ ,$$

where $\tilde{\eta}$ is the hybrid average elasticity $\tilde{\eta} \equiv \frac{1}{X} \sum_k x_k(p^0) \eta_k(p^*)$

- So $MR^D \geq MR^U$ if $\tilde{\eta} \geq \eta(p^0)$
- With isoelastic demands $\tilde{\eta} = \eta(p^0)$.

Conclusions

- Aim has been to simplify and generalize – interdependent demands and n products – analysis of the output effect of third-degree price discrimination
- Average output differences can sometimes be related to endpoint profit differences
- When all products are supplied the effect can be signed when $\eta(p^*)$ is not between $\beta(p^*)$ and $\eta(p^0)$
- Complementary approaches extend the analysis, as illustrated by isoelastic demands.